



**UNIMORE**  
UNIVERSITÀ DEGLI STUDI DI  
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria  
“Enzo Ferrari”

# **Fondamenti di elettronica Esperienza di laboratorio: Esempio di utilizzo del protocollo seriale I2C - Magnetometro**

Dott. Tommaso Zanotti

email: [tommaso.zanotti@unimore.it](mailto:tommaso.zanotti@unimore.it)

# Il magnetometro LIS3MDL



LGA-12L  
(2.0 x 2.0 x 1.0 mm)

## Features

- Wide supply voltage, 1.9 V to 3.6 V
- Independent IO supply (1.8 V)
- $\pm 4/\pm 8/\pm 12/\pm 16$  gauss selectable magnetic full scales
- Continuous and single-conversion modes
- 16-bit data output
- Interrupt generator
- Self-test
- I<sup>2</sup>C/SPI digital output interface
- Power-down mode / low-power mode
- ECOPACK and RoHS compliant

## Applications

- Magnetometers
- Compasses

## Description

The LIS3MDL is an ultralow-power high-performance 3-axis magnetic sensor with user-selectable full scales of  $\pm 4/\pm 8/\pm 12/\pm 16$  gauss.

The self-test capability allows the user to check the functioning of the sensor in the final application.

The device may be configured to generate interrupt signals for magnetic field detection.

The LIS3MDL includes an I<sup>2</sup>C serial bus interface that supports standard and fast mode (100 kHz and 400 kHz) and SPI serial standard interfaces.

The device is available in a small thin plastic, land grid array (LGA) package and is guaranteed to operate over an extended temperature range of -40°C to +85°C.

Per maggiori dettagli fare riferimento al datasheet nella cartella  
*"datasheets/MEMS/lis3mdl.pdf"*

### Product status link

[LIS3MDL](#)

### Product summary

|                        |               |
|------------------------|---------------|
| Order code             | LIS3MDLTR     |
| Temperature range [°C] | -40 to +85    |
| Package                | LGA-12L       |
| Packing                | Tape and reel |

### Product resources

[AN4602](#) (device application note)

[TN0018](#) (design and soldering)

# Il magnetometro LIS3MDL

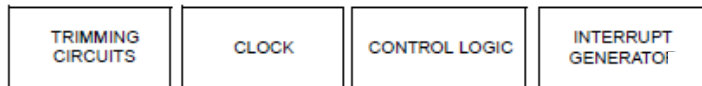
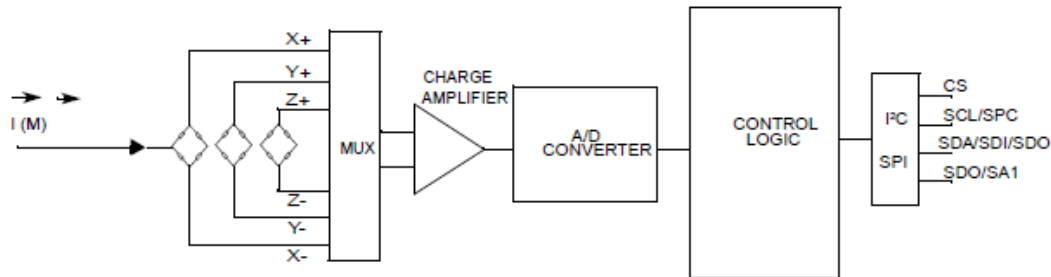
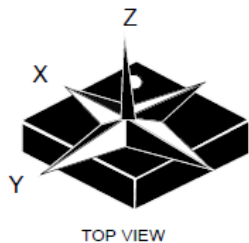
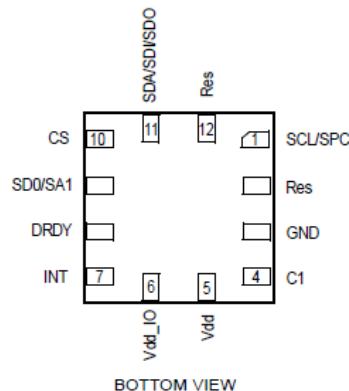


Table 1. Pin description

| Pin# | Name              | Function                                                                                                                                                                   |
|------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | SCL<br>SPC        | I <sup>2</sup> C serial clock (SCL)<br>SPI serial port clock (SPC)                                                                                                         |
| 2    | Reserved          | Connect to GND                                                                                                                                                             |
| 3    | GND               | Connect to GND                                                                                                                                                             |
| 4    | C1                | Capacitor connection (C1 = 100 nF)                                                                                                                                         |
| 5    | Vdd               | Power supply                                                                                                                                                               |
| 6    | Vdd_IO            | Power supply for I/O pins                                                                                                                                                  |
| 7    | INT               | Interrupt                                                                                                                                                                  |
| 8    | DRDY              | Data ready                                                                                                                                                                 |
| 9    | SDO<br>SA1        | SPI serial data output (SDO)<br>I <sup>2</sup> C less significant bit of the device address (SA1)                                                                          |
| 10   | CS                | Enable SPI<br>I <sup>2</sup> C/SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C disabled) |
| 11   | SDA<br>SDI<br>SDO | I <sup>2</sup> C serial data (SDA)<br>SPI serial data input (SDI)<br>3-wire interface serial data output (SDO)                                                             |
| 12   | Reserved          | Connect to GND                                                                                                                                                             |



DIRECTION OF  
DETECTABLE  
MAGNETIC FIELDS



# Il magnetometro LIS3MDL

Table 2. Magnetic characteristics

| Symbol | Parameter                   | Test conditions                                                | Min. | Typ. <sup>(1)</sup> | Max. | Unit          |
|--------|-----------------------------|----------------------------------------------------------------|------|---------------------|------|---------------|
| FS     | Measurement range           |                                                                |      | ±4                  |      | gauss         |
|        |                             |                                                                |      | ±8                  |      |               |
|        |                             |                                                                |      | ±12                 |      |               |
|        |                             |                                                                |      | ±16                 |      |               |
| GN     | Sensitivity                 | FS = ±4 gauss                                                  |      | 6842                |      | LSB/<br>gauss |
|        |                             | FS = ±8 gauss                                                  |      | 3421                |      |               |
|        |                             | FS = ±12 gauss                                                 |      | 2281                |      |               |
|        |                             | FS = ±16 gauss                                                 |      | 1711                |      |               |
| Zgauss | Zero-gauss level            | FS = ±4 gauss                                                  |      | ±1                  |      | gauss         |
| RMS    | RMS noise                   | X-axis; FS = ±12 gauss;<br>Ultrahigh-performance mode          |      | 3.2                 |      | mgauss        |
|        |                             | Y-axis; FS = ±12 gauss<br>Ultrahigh-performance mode           |      | 3.2                 |      | mgauss        |
|        |                             | Z-axis; FS = ±12 gauss<br>Ultrahigh-performance mode           |      | 4.1                 |      | mgauss        |
|        |                             |                                                                |      |                     |      |               |
| NL     | Nonlinearity                | Best-fit straight line<br>FS = ±12 gauss<br>Applied = ±6 gauss |      | ±0.12               |      | %FS           |
| ST     | Self test <sup>(2)</sup>    | X-axis<br>FS = ±12 gauss                                       | 1    |                     | 3    | gauss         |
|        |                             | Y-axis<br>FS = ±12 gauss                                       | 1    |                     | 3    |               |
|        |                             | Z-axis<br>FS = ±12 gauss                                       | 0.1  |                     | 1    |               |
|        |                             |                                                                |      |                     |      |               |
| DF     | Magnetic disturbance field  | Zero-gauss offset starts to degrade                            |      |                     | 50   | gauss         |
| Top    | Operating temperature range |                                                                | -40  |                     | +85  | °C            |

1. Typical specifications are not guaranteed.

2. Absolute value

L'intensità del campo geomagnetico sulla superficie terrestre varia da 25 a 65 microtesla (o da 0.25 a 0.65 gauss).

# II magnetometro LIS3MDL

## I<sup>2</sup>C interface

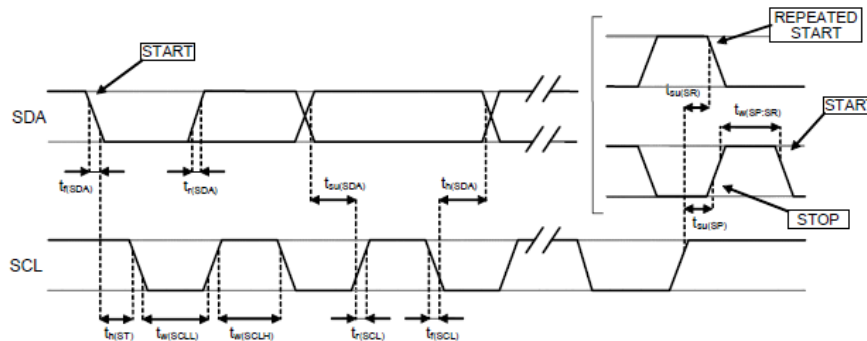
Table 6. I<sup>2</sup>C slave timing values

| Symbol              | Parameter                                      | I <sup>2</sup> C standard mode <sup>(1)</sup> |      | I <sup>2</sup> C fast mode <sup>(1)</sup> |      | Unit    |
|---------------------|------------------------------------------------|-----------------------------------------------|------|-------------------------------------------|------|---------|
|                     |                                                | Min.                                          | Max. | Min.                                      | Max. |         |
| $f_{(SCL)}$         | SCL clock frequency                            | 0                                             | 100  | 0                                         | 400  | kHz     |
| $t_{W(SCLL)}$       | SCL clock low time                             | 4.7                                           |      | 1.3                                       |      | $\mu$ s |
| $t_{W(SCLH)}$       | SCL clock high time                            | 4.0                                           |      | 0.6                                       |      |         |
| $t_{su(SDA)}$       | SDA setup time                                 | 250                                           |      | 100                                       |      | ns      |
| $t_h(SDA)$          | SDA data hold time                             | 0                                             | 3.45 | 0                                         | 0.9  | $\mu$ s |
| $t_r(SDA) t_r(SCL)$ | SDA and SCL rise time                          |                                               | 1000 | $20 + 0.1C_b^{(2)}$                       | 300  | ns      |
| $t_f(SDA) t_f(SCL)$ | SDA and SCL fall time                          |                                               | 300  | $20 + 0.1C_b^{(2)}$                       | 300  |         |
| $t_h(ST)$           | START condition hold time                      | 4                                             |      | 0.6                                       |      | $\mu$ s |
| $t_{su(SR)}$        | Repeated START condition setup time            | 4.7                                           |      | 0.6                                       |      |         |
| $t_{su(SP)}$        | STOP condition setup time                      | 4                                             |      | 0.6                                       |      |         |
| $t_{W(SP:SR)}$      | Bus free time between STOP and START condition | 4.7                                           |      | 1.3                                       |      |         |

1. Data based on standard I<sup>2</sup>C protocol requirement, not tested in production.

2.  $C_b$  = total capacitance of one bus line, in pF.

Figure 4. I<sup>2</sup>C slave timing diagram



The serial interfaces are mapped to the same pads. To select/exploit the I<sup>2</sup>C interface, the **CS line must be tied high** (that is, connected to Vdd\_IO).

The LIS3MDL I<sup>2</sup>C is a bus slave. The I<sup>2</sup>C is employed to write data to registers whose content can also be read back.

**The slave address (SAD)** associated to the LIS3MDL is **00111x0b**, whereas the x bit is modified by the SDO/SA1 pin in order to modify the device address. If the SDO/SA1 pin is connected to the voltage supply, the address is 0011110b, otherwise, if the SDO/SA1 pin is connected to ground, the address is 0011100b.

**Come troviamo l'indirizzo da utilizzare per la nostra scheda?**

# Il magnetometro LIS3MDL

## I<sup>2</sup>C interface

**Table 11. Transfer when master is writing one byte to slave**

|        |    |         |     |     |     |      |     |    |
|--------|----|---------|-----|-----|-----|------|-----|----|
| Master | ST | SAD + W |     | SUB |     | DATA |     | SP |
| Slave  |    |         | SAK |     | SAK |      | SAK |    |

**Table 12. Transfer when master is writing multiple bytes to slave**

|        |    |         |     |     |     |      |     |      |     |    |
|--------|----|---------|-----|-----|-----|------|-----|------|-----|----|
| Master | ST | SAD + W |     | SUB |     | DATA |     | DATA |     | SP |
| Slave  |    |         | SAK |     | SAK |      | SAK |      | SAK |    |

**Table 13. Transfer when master is receiving (reading) one byte of data from slave**

|        |    |         |     |     |     |    |         |     |      |      |    |
|--------|----|---------|-----|-----|-----|----|---------|-----|------|------|----|
| Master | ST | SAD + W |     | SUB |     | SR | SAD + R |     |      | NMAK | SP |
| Slave  |    |         | SAK |     | SAK |    |         | SAK | DATA |      |    |

**Table 14. Transfer when master is receiving (reading) multiple bytes of data from slave**

|        |    |           |     |     |     |    |           |     |      |     |      |     |      |      |    |
|--------|----|-----------|-----|-----|-----|----|-----------|-----|------|-----|------|-----|------|------|----|
| Master | ST | SAD+<br>W |     | SUB |     | SR | SAD+<br>R |     |      | MAK |      | MAK |      | NMAK | SP |
| Slave  |    |           | SAK |     | SAK |    |           | SAK | DATA |     | DATA |     | DATA |      |    |

SAK = Slave ACK

Data are transmitted in byte format (DATA). Each data transfer contains **8 bits**. The number of bytes transferred per transfer is unlimited. Data is transferred with the **most significant bit (MSb) first**.

# Configurazione del sensore

Table 15. Register address map

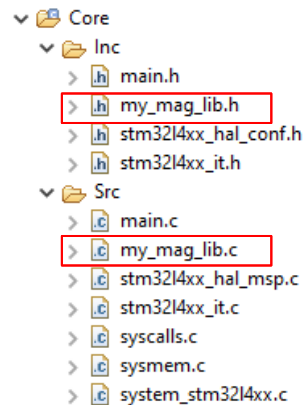
| Name             | Type | Register address |           | Default  | Comment             |
|------------------|------|------------------|-----------|----------|---------------------|
|                  |      | Hex              | Binary    |          |                     |
| Reserved         |      | 00 - 04          | --        | --       | Reserved            |
| OFFSET_X_REG_L_M | R/W  | 05               | 00000101  | 00000000 | Hard-iron registers |
| OFFSET_X_REG_H_M | R/W  | 06               | 00000110  | 00000000 |                     |
| OFFSET_Y_REG_L_M | R/W  | 07               | 00000111  | 00000000 |                     |
| OFFSET_Y_REG_H_M | R/W  | 08               | 00001000  | 00000000 |                     |
| OFFSET_Z_REG_L_M | R/W  | 09               | 00001001  | 00000000 |                     |
| OFFSET_Z_REG_H_M | R/W  | 0A               | 00001010  | 00000000 |                     |
| Reserved         |      | 0B - 0E          | --        | --       | Reserved            |
| WHO_AM_I         | R    | 0F               | 0000 1111 | 00111101 | Dummy register      |
| Reserved         |      | 10 - 1F          | --        | --       | Reserved            |
| CTRL_REG1        | R/W  | 20               | 0010 0000 | 00010000 |                     |
| CTRL_REG2        | R/W  | 21               | 0010 0001 | 00000000 |                     |
| CTRL_REG3        | R/W  | 22               | 0010 0010 | 00000011 |                     |
| CTRL_REG4        | R/W  | 23               | 0010 0011 | 00000000 |                     |
| CTRL_REG5        | R/W  | 24               | 0010 0100 | 00000000 |                     |
| Reserved         |      | 25 - 26          | --        | --       | Reserved            |
| STATUS_REG       | R    | 27               | 0010 0111 | Output   |                     |
| OUT_X_L          | R    | 28               | 0010 1000 | Output   |                     |
| OUT_X_H          | R    | 29               | 0010 1001 | Output   |                     |
| OUT_Y_L          | R    | 2A               | 0010 1010 | Output   |                     |
| OUT_Y_H          | R    | 2B               | 0010 1011 | Output   |                     |
| OUT_Z_L          | R    | 2C               | 0010 1100 | Output   |                     |
| OUT_Z_H          | R    | 2D               | 0010 1101 | Output   |                     |
| TEMP_OUT_L       | R    | 2E               | 0010 1110 | Output   |                     |
| TEMP_OUT_H       | R    | 2F               | 0010 1111 | Output   |                     |
| INT_CFG          | R/W  | 30               | 00110000  | 11101000 |                     |
| INT_SRC          | R    | 31               | 00110001  | 00000000 |                     |
| INT_THS_L        | R/W  | 32               | 00110010  | 00000000 |                     |
| INT_THS_H        | R/W  | 33               | 00110011  | 00000000 |                     |

Per configurare il sensore occorre scrivere dei valori all'interno di specifici registri.

Fare riferimento al datasheet per conoscere il significato di ciascun bit all'interno dei vari registri

# Steps per la creazione di un progetto di esempio

1. Creare un nuovo progetto dove abilitiamo le periferiche I2C2 e USART1
2. Creare una cartella sotto Drivers in cui creare i file .c e .h della libreria and add the folder to the compiler path.



3. Definire gli indirizzi dei registri ed implementare le funzioni per inizializzare il sensore e leggere i dati
4. Istanziare le nuove funzioni nel main.c e verificarne il corretto funzionamento